

```
KJagtap@cqlsh> DESCRIBE KEYSPACES ;

operations          safelite_poc        dm_accident_fdb     txn_mvr
ml_price_fdb        system              subudhid             telematics
ase                 telematics_ingestion ref_core             dm_telematics
txn_fuel_fdb        odometer            dm_accident          txn_bill
system_auth         alt_fund_fdb        txn_titleandreg      dse_system
perfdashboard_fdb   geotab_integration filodb               agg_fdb
txn_maintenance     dba_test1          job_test_fdb
keyspace1           telematics_messages etl_metadata         txn_inv_fdb
alt_fund            test               dse_security
txn_remarketing     "HiveMetaStore"    txn_personaluse_fdb
telematics_solr     agg_dc2            txn_inv
core_fdb            subudhid123        spectra
telematics_bkp      loadtest           cfs_archive
dm_finance          core               txn_acdt_fdb
cfs                 advanced_analytics system_traces
txn_txn_fdb         dm_finance_fdb     txn_fuel
advanced_analytics_fdb txn_acdt           rest_api_meta
txn_titleandreg_fdb txn_dvcr           txn_personaluse
txn_maintenance_fdb core_txn_stg       txn_telematics_fdb
txn_billing         filodb_admin       api_log_monitor
ref_core_fdb        txn_dvcr_fdb       dev
txn_bill_fdb        txn_mvr_fdb        telematics_stg
txn_remarketing_fdb dm_telematics_fdb  fdb_txn_remarketing
odometer_fdb        txn_telematics

KJagtap@cqlsh>
```

Figure 5: The *describe keyspace* command

```
KJagtap@cqlsh> ALTER KEYSPACE cassandradb WITH replication = ('class': 'SimpleStrategy', 'replication_factor':2) AND durable_writes = false;
KJagtap@cqlsh> desc keyspace cassandradb ;

CREATE KEYSPACE cassandradb WITH replication = ('class': 'SimpleStrategy', 'replication_factor': '2') AND durable_writes = false;
```

Figure 6: *Alter keyspace* command

```
KJagtap@cqlsh> drop keyspace cassandradb ;
KJagtap@cqlsh> desc keyspace cassandradb ;

Keyspace 'cassandradb' not found.
KJagtap@cqlsh>
KJagtap@cqlsh>
```

Figure 7: *Drop keyspace* command

Keyspace: This is a name space in Cassandra that defines data replication on nodes. A cluster contains one keyspace per node. While creating a keyspace, we should keep in mind the points given below.

Data replication: Cassandra provides a strategy to determine the nodes where replicas (copies) are placed. So in case any node goes down, your data remains safe and available since there are copies on other nodes as well. This is the advantage of using Cassandra.

Replication factor: This is the number of replicas in the cluster that

will receive copies of the same data. If the *Replication_factor* = 2, it means data will be replicated to two different nodes.

SimpleStrategy: This is the process by which replicas are placed in the ring. It is used when we have only one data centre. As a best practice, we should not use this strategy in a production environment.

NetworkTopologyStrategy:

This strategy is used when we have multiple data centres, and can be used in a production environment.

Durable_writes: Another keyspace option is *durable_writes*, and by default, it is always set to 'true'. If it's set to 'false', then no updates will be written to the commit log.

Here is an example of the 'create keyspace' command:

```
CREATE KEYSPACE Keyspace cassandradb
WITH replication = {'class':
NetworkTopologyStrategy, 'Analytics':
1} And durable_writes = true;
```

- In Figure 4, *Cassandradb* is used as *keyspace_name*.
- *NetworkTopologyStrategy* is used as the replication strategy.
- *Analytics* is used as *replication_factor* name.

Now the *Cassandradb* keyspace is created and we can use the 'describe keyspace' command to see keyspace as shown in Figure 5.

Here is an example of the 'alter keyspace' command:

```
ALTER KEYSPACE cassandradb
WITH replication = {'class':
'SimpleStrategy', 'replication_factor':2} AND durable_writes =
false;
```

This keyspace is explained in 'command and short description section'.

Figure 7 gives an example of the 'drop keyspace' command in Cassandra.

Now that you have got all the basic information on this database, you are ready to explore it further on your own. Do try it out; you will find it has a range of useful features. **END** 

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